Weichen Li

Ph.D. Candidate in Machine Learning

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Research Profile

Final-year Ph.D. candidate in the Machine Learning Group at RPTU Kaiserslautern-Landau (advisor: Prof. Sophie Fellenz). My research explores how AI systems can make decisions when human values are plural, uncertain, or conflicting. I work on *pluralistic AI alignment*, *multi-objective and safe reinforcement learning*, and *moral decision-making in language-based agents*, with a particular interest in connecting normative questions about human values to practical AI decision-making methods.

Focus areas Pluralistic alignment · Inference-time value alignment · Offline Safe RL · Multi-objective RL

Education

Ph.D., Computer Science, *RPTU University Kaiserslautern-Landau*, Germany
Thesis: *Value-Aligned Reinforcement Learning: From Language-Based to Multi-Objective Decision Making*

M.Sc., Computational Linguistics, *Ludwig Maximilian University of Munich*, Germany

B.A., Sociology (minor: Computer Science), *University of Bamberg*, Germany

Selected Publications

Pluralistic & Multi-Objective AI Alignment

NeurIPS 2026 **Pluralistic AI Alignment Requires Inference-Time Multi-Objective Control**
W. Li, M. Stojanović, D. Neider, M. Kloft, S. Fellenz
Advances in Neural Information Processing Systems (Position Paper) · To appear. Earlier version at ICML 2026 Pluralistic Alignment Workshop: `https://openreview.net/forum?id=utmfkj6dOY`

AAAI 2026 **TORA: Train Once, Realign Anytime for Offline Multi-Objective RL**
W. Li, W. Mustafa, M. Monteiro, P. Wang, M. Kloft, S. Fellenz
Proceedings of the AAAI Conference on Artificial Intelligence · `https://doi.org/10.1609/aaai.v40i44.41095`

ICML 2026 **Landmark-Guided Policy Optimization for Multi-Objective Language Model Selection**
M. Monteiro, **W. Li**, P. Wang, M. Kloft, S. Fellenz
International Conference on Machine Learning · `https://openreview.net/forum?id=fexmo79Kxo`

Ethical & Value-Aligned Decision-Making

AISTATS 2024 **Ethics in Action: Training RL Agents for Moral Decision-Making in Text-Based Adventure Games**
W. Li, R. Devidze, W. Mustafa, S. Fellenz
Int. Conference on Artificial Intelligence and Statistics · `https://proceedings.mlr.press/v238/li24i.html`

EMNLP-W 2025 **Inference-Time Value Alignment in Offline RL: Leveraging LLMs for Reward and Ethical Guidance**
W. Li, W. Mustafa, R. Devidze, M. Kloft, S. Fellenz
WORDPLAY Workshop @ EMNLP · `https://wordplay-workshop.github.io/pdfs/24.pdf`

ECML-W 2025 **Inference-Time Preference-Aligned Diffusion Planning for Safe Offline RL**
W. Li, W. Mustafa, P. Wang, M. Kloft, S. Fellenz
HHMLDM Workshop @ ECML-PKDD · **Oral presentation**

Language-Based Agents & Reasoning

ECML-PKDD 2023 **Learning to Play Text-Based Adventure Games with Maximum Entropy RL**
W. Li, R. Devidze, S. Fellenz
European Conference on Machine Learning and Knowledge Discovery in Databases · https://link.springer.com/chapter/10.1007/978-3-031-43421-1_3

IEEE ICBK 2021 **Topic-Guided Knowledge Graph Construction for Argument Mining**
W. Li, P. Abels, Z. Ahmadi, S. Burkhardt, B. Schiller, I. Gurevych, S. Kramer
IEEE International Conference on Big Knowledge · https://doi.org/10.1109/ICKG52313.2021.00049

Medical AI Experience

Medical Augmented Reality Summer School 2026. Selected participant in a summer school organized by the Technical University of Munich (TUM), focusing on AI, robotics, and extended reality (XR) for clinical applications.

Hackathon Project. *Spatial AR Visualization of AI Uncertainty in Surgical Environments*. Awarded Best Innovation Project.

Teaching & Supervision

Supervision Bachelor's and Master's projects on LLMs for SMILES/SMARTS molecular generation; RL for molecular property prediction and thermodynamic group-contribution methods; retrieval-augmented (RAG) chatbots for university library services.

Teaching Teaching assistant, *Neural Networks for NLP*.